18-07-2026

Team will upload wednesday class recording soon. Sorry for the day

please read this

Agenda: ML-20

Gradient Boosting } Regression
Xgboost
practicals + docker intro

$$m1 \xrightarrow{\text{errors}} m2 \xrightarrow{\text{error}} m3 \rightarrow \dots 100$$

Boosting

Gradient Boosting

lr

technique

- sequentials models
  - DT → simple models
  - → stumps [1, 3, 6, 8]

M1 M2 M3

→ → 0.1 x → 0.1 x ← lr

$$\text{loss function} = \frac{1}{2}(y - \hat{y})^2$$

Dataset

ID	Size	Beds	price (y)
1	800	2	120
2	900	2	130
3	1000	3	150
4	1100	3	170
5	1200	4	200

$lr = 0.1$

Data (X)

↓

model [F0]

↓

154

Step: 0 → Initial model (F0)

$F0 = \text{mean}(y) \rightarrow (120+130+150+170+200)/5 = 154$

→ calculate residuals $(y-F) = (y-F_0)$

ID	Size	Beds	price (y)	r1 $(y-F_0)$
1	800	2	120	-34
2	900	2	130	-24
3	1000	3	150	-4
4	1100	3	170	+16
5	1200	4	200	+46

↑

target for the stump

GB (Regression)

$\text{loss} = \frac{1}{2}(y-F(x))^2$

gradient (first derivate)

$\rightarrow g = \frac{\partial L}{\partial F} = F - y$

changing direction to lower the loss →

$-g = y - F$

→ candidate splits

ID	Size	Beds	price (y)	r1 $(y-F_0)$
1	800	2	120	-34
2	900	2	130	-24
3	1000	3	150	-4
4	1100	3	170	+16
5	1200	4	200	+46

850, 950, 1050, 1150

2.5, 3.5

for i in $\boxed{\text{candidates}}$:

→ left (A) right (B)

→ mean (A) mean (B) ← residual mean

→ SSE (residual) over both sides $(r - \text{mean})^2$

Feature	Threshold	left (A)	Right (B)	mean (A)	mean (B)	SSE
Size	1050	[1,2,3]	[4,5]	-20.66	31	916 ← lowest
Size	950	[1,2]	[3,4,5]	-29	19.33	1316.66
Size	1150	[1,2,3,4]	[5]	-11.5	46	1475
Size	850	[1]	[2,3,4,5]	-34	8.5	2675
Beds	2.5	[1,2]	[3,4,5]	-29	19.33	1316.66
Beds	3.5	[1,2,3,4]	[5]	-11.5	46	1475

Full calculation example below

Size < 1050 chosen as stump

left

ID	size	residuals
1	800	-34
2	900	-24
3	1000	-4

mean = -20.66

right

ID	size	residuals
4	1100	16
5	1200	46

mean = 31 ← from above table

SSE(A) =

$(-34 - (-20.66)) = -13.33 \rightarrow$ squared $= 177$

$(-24 - (-20.66)) = -3.33 \rightarrow$ $= 11.11$

$(-4 - (-20.66)) = 16.66 \rightarrow$ $= 277.77$

466

SSE(B)

$(16 - 31) = -15 \rightarrow$ squared $= 225$

$(46 - 31) = 15 \rightarrow 225$

450

Total SSE = SSE(A) + SSE(B)

= 916.66

$\boxed{F_0(x) \rightarrow 154}$ ← model-1

$h_1(x) \rightarrow$ Size ≤ 1050 ← model-2

↙ -20.66 ↘ 31

$F1(x) = F0 + lr * h1(x)$

ID	Size	Beds	price (y)	r1 (y - F0)	F1	r2 (y - F1)
1	800	2	120	-34	151.93	-31.93
2	900	2	130	-24	151.93	-21.93
3	1000	3	150	-4	151.93	-1.93
4	1100	3	170	+16	157.1	12.90
5	1200	4	200	+46	157.1	42.90

compute F1:

ID-1 (A) : $\underbrace{154}_{F0} + 0.1 \times (-20.66) = 151.93$

800 ≤ 1050

-20.66 31.0

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we will use r2 as the next stump

Feature	Threshold	left (A)	Right (B)	mean (A)	mean (B)	SSE
Size	1050	[1,2,3]	[4,5]			
Size	950	[1,2]	[3,4,5]			
Size	1150	[1,2,3,4]	[5]			
Size	850	[1]	[2,3,4,5]			
Beds	2.5	[1,2]	[3,4,5]			
Beds	3.5	[1,2,3,4]	[5]			

Feature	Threshold	left (A)	Right (B)	mean (A)	mean (B)	SSE
Size	1050	[1,2,3]	[4,5]	-18.60	27.90	916.66 ← lowest
Size	950	[1,2]	[3,4,5]	-26.93	17.95	1093.35
Size	1150	[1,2,3,4]	[5]	-10.72	42.9	1210.85
Size	850	[1]	[2,3,4,5]	-31.93	8.5	2236.69
Beds	2.5	[1,2]	[3,4,5]	-26.93	17.95	1093.35
Beds	3.5	[1,2,3,4]	[5]	-10.72	42.9	1210.85

$$h_2(x) \longrightarrow \text{size} \leq 1050$$

$\swarrow$ −18.6 $\qquad \searrow$ 27.9

→ new stump

→ model-3

$$F_1(x) = f_0 + lr * h_1(x)$$

$$F_2(x) = f_0 + lr * h_1(x) + lr * h_2(x)$$

$$= F_1(x) + lr * h_2(x)$$

copied from above for F1 value

ID	Size	Beds	price (y)	r1 (y-f0)	F1	r2 (y-F1)
1	800	2	120	-34	151.93	-31.93
2	900	2	130	-24	151.93	-21.93
3	1000	3	150	-4	151.93	-1.93
4	1100	3	170	+16	157.1	12.90
5	1200	4	200	+46	157.1	42.90

ID	Size	Beds	price (y)	r1 (y-f0)	F1	r2 (y-F1)	F2	r3 (y-F2)
1	800	2	120	-34	151.93	-31.93	150.07	-30.07
2	900	2	130	-24	151.93	-21.93	150.07	-20.07
3	1000	3	150	-4	151.93	-1.93	150.07	-0.07
4	1100	3	170	+16	157.1	12.90	159.89	+10.11
5	1200	4	200	+46	157.1	42.90	159.89	+40.11

compute - F2

ID $= f_1(x) + lr * h_2(x)$

[1,2,3] $= 151.93 + 0.1 * (-18.6) = 150.07$

ID

[4,5] $= 157.1 + 0.1 * 27.9 = 159.89$

Because the residuals maintain the same sign pattern (IDs 1–3 negative, 4–5 positive), splitting along Size still best separates negative vs positive residuals and thus minimizes SSE. As iterations progress residual magnitudes shrink, and at some point another split (e.g., Beds or a different Size cutoff) might become best, but early on Size is the dominant predictive feature.

Xgboost

Xtreme - gradient boost

Dataset

↓ ← F0 → mean(target)

for each data-point:

gradient, hessian

for each split:

L

R

total → G(left), H(left) → G_L, H_L

G(Right), H(Right) G_R, H_R

compute gain = Score(L) + Score(R) - Score(Parent) - γ

$$\text{score} = \frac{-G^2}{H + \lambda}$$

choose-split with highest possible gain

For each leaf compute final leaf weight

w_L

w_R

$$\text{Final prediction } (F_{t+1}) = F_0 + \eta \cdot f_t(x)$$

gB ← xgb

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loss = squared error

$$l = \frac{1}{2}(y - \hat{y})^2$$

$$\text{gradient} = \frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

$$\text{hessian} = \frac{\partial^2 L}{\partial \hat{y}^2} = 1$$

λ (regularization) = 1

γ (minimum gain to split) = 0

λ → denominator (L2)
→ shrinks score

γ → substraction penalty (L1)

α → leaf weight calc, not split score (lr)

$$w = \frac{-G}{H + \lambda}$$

$\eta = 1$
for simple calculation

ID	Size	Beds	price (y)	$F_0(\hat{y}_i^{(0)})$	$g = \hat{y}_0 - y$	h
1	800	2	120	154	34	1
2	900	2	130	154	24	1
3	1000	3	150	154	4	1
4	1100	3	170	154	-16	1
5	1200	4	200	154	-46	1

Iteration-1 → Build Tree

		Gain
Size	850	1156
Size	950	**1962.33**
Size	1050	707
Beds	2.5	1156
Beds	3.5	707

highest gain → Calculation

Size < 950

L: [1, 2] R: [3, 4, 5]

$g = 34 + 24 = 58$, $h = 2$

$g = 4 - 16 - 46 = -58$, $h = 3$

$$\text{score}(L) = -\frac{58^2}{2+1} = -1121.33 \qquad \text{score}(R) = -\frac{(-58)^2}{3+1} = -841$$

score (Parent)

$g = 34 + 24 + 4 - 16 - 46 = 0$

$h = 5$

$$= -\frac{0^2}{5+1} = 0$$

Gain $= 1121.33 + 841 =$ **1962.33**

$$(F_{t+1}) = F_0 + \eta \cdot f_t(x)$$

leaf-weights:

$$w = -\frac{G}{H + \lambda}$$

Size < 950

[1,2] [3,4,5]

$$w_L = -\frac{58}{2+1} = -19.33 \qquad w_R = -\frac{-58}{3+1} = 14.5$$

ID	Size	Beds	price (y)	$F_0(\hat{y}_i^{(0)})$	leaf	weight	F_1
1	800	2	120	154	L	-19.33	134.67
2	900	2	130	154	L	-19.33	134.67
3	1000	3	150	154	R	+14.5	168.5
4	1100	3	170	154	R	+14.5	168.5
5	1200	4	200	154	R	+14.5	168.5

ID-1 → $154 + 1 * (-19.33) = 134$

Iteration-2

ID	Size	Beds	price (y)	F_1	g	h
1	800	2	120	134.67	14.67	1
2	900	2	130	134.67	4.67	1
3	1000	3	150	168.5	18.5	1
4	1100	3	170	168.5	-1.5	1
5	1200	4	200	168.5	-31.5	1

	Gradient-Boost	Xgboost
Uses derivative of loss	Yes	Yes
Uses second order derivative	NO	Yes
split score	SSE	Based on gain (g, h, λ)
Regularization	weak (α)	strong $(\lambda, \alpha, \gamma)$
Tree complexity control	max-depth	prunes automatically + max-depth
leaf value	mean residual	$-\frac{G}{H+\lambda}$
System speed	slower	much faster, parallel